# Natural language processing

## Text as data: how machines read

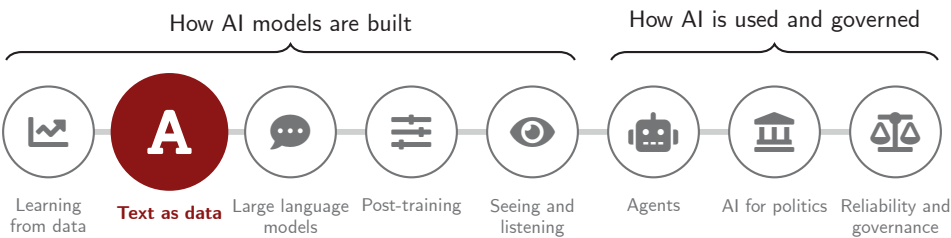
POLI 158: AI & Politics · Session 5

Yiqing Xu

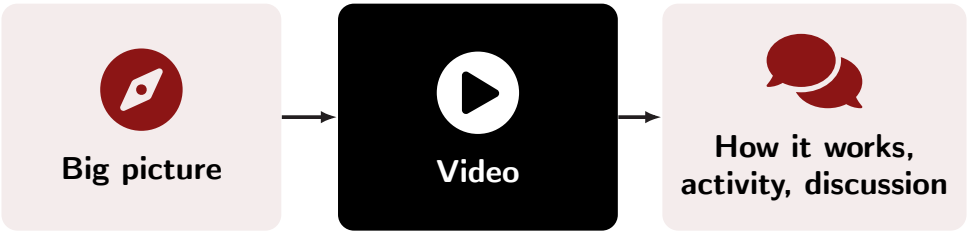
**TODAY'S QUESTION**

If an AI learned only from censored text, what would it believe, and could you tell?

# The eight Wednesday sessions



# Today in three parts



# Where computers read text

## Everyday life


autocomplete


translation


search


chatbots

## Research


sorting posts by stance


placing parties left to right


finding what censors delete

Grimmer and Stewart (2013); King, Pan, and Roberts (2013)

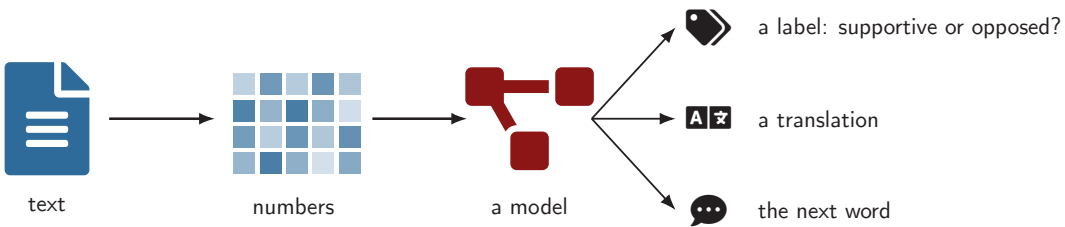
## Politics and policy


screening hate speech

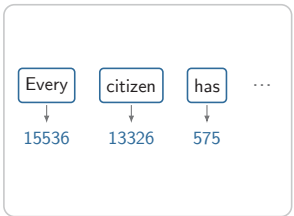

voting advice


sorting public comments

# From text to numbers

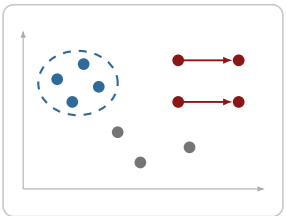


# Three concepts in today's video



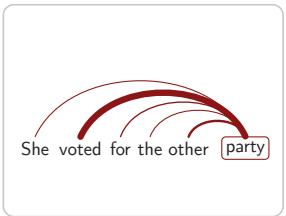

**Tokens**

text becomes numbers



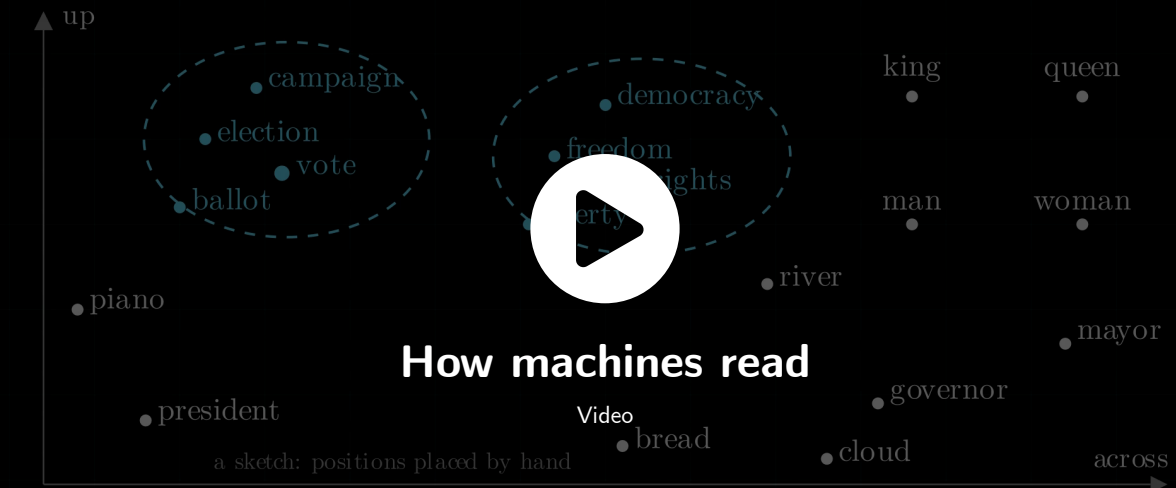
**Embeddings**

a map of meaning




**Attention**

context changes meaning




## How machines read

Video

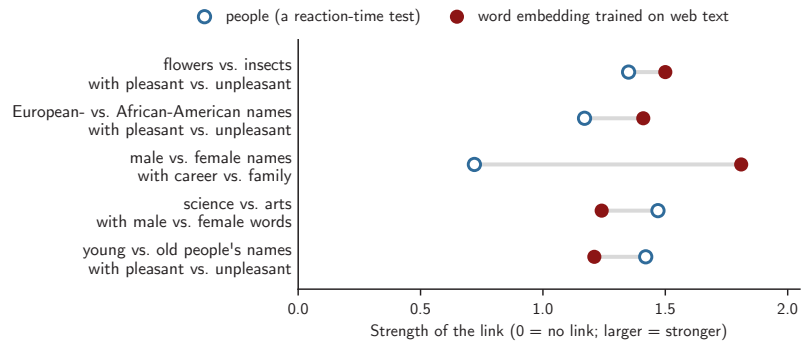
# One sentence in five languages

| Language | Universal Declaration of Human Rights, Article 1, first sentence | Tokens |
|---|---|---|
| English | All human beings are born free and equal in dignity and rights. | ? |
| Chinese | 人人生而自由，在尊严和权利上一律平等。 | ? |
| Hindi | सभी मनुष्यों को गौरव और अधिकारों के मामले में जन्मजात स्वतन्त्रता और समानता प्राप्त है । | ? |
| Burmese | လူတိုင်းသည် တူညီ လွတ်လပ်သော ဂုဏ်သိက္ခာဖြင့် လည်းကောင်း၊ တူညီလွတ်လပ်သော အခွင့်အရေးများဖြင့် လည်းကောင်း၊ မွေးဖွားလာသူများ ဖြစ်သည်။ | ? |
| Amharic | የሰው፡ልጅ፡ሁሉ፡ሲወለድ፡ነጻና፡በክብርና፡በመብትም፡እኩል፡ነት፡ያለው፡ነው።። | ? |

Guess which line needs the most tokens. Then try platform.openai.com/tokenizer.

Source: United Nations (1948), Universal Declaration of Human Rights. English and Chinese: official UN texts. Hindi, Burmese, Amharic: translations from the UN human rights office's translation project, as encoded by UDHR in Unicode. Petrov et al. (2023).

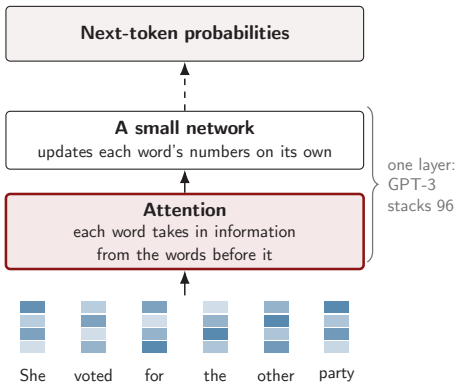
# Stereotypes in word embeddings




People: how quickly they pair two kinds of words (the Implicit Association Test).
Embedding: how close the two kinds of words sit on the map.

Source: Caliskan, Bryson, and Narayanan (2017), *Science*, Table 1 (five of its ten tests). The authors do not claim the two sets of numbers are directly comparable: read the directions, not the gaps.

# The transformer




- GPT: generative pre-trained transformer.

Source: Vaswani et al. (2017); Brown et al. (2020), Table 2.1; 3Blue1Brown (2024); the 2025 course slides.

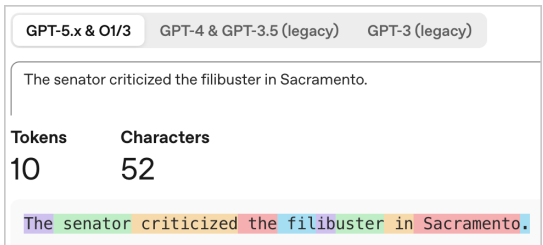
# Activity: tokens and word embeddings

# Activity, part 1: tokens

In pairs, one laptop per pair. No login needed.

## platform.openai.com/tokenizer

1. Type a sentence in another language you know, then in English. Which needs more tokens?
2. Click the older tab, "GPT-4 & GPT-3.5 (legacy)." Does the gap grow or shrink?
3. Type your name and your hometown. Which break into pieces?
4. Type "strawberry." Why might a chatbot miscount its r's?




Example: 10 tokens. "Filibuster" breaks into three pieces.

Screenshot: OpenAI's tokenizer page, default tab, October 6, 2026.

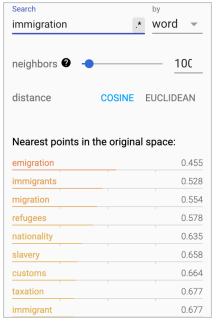
# Activity, part 2: word embeddings

Same pairs. The page takes a few seconds to load its map.

projector.tensorflow.org

1. Type "immigration" in the search box on the right. Read its nearest words.
2. Search two political words of your own.
3. Find a neighbor that is not part of the word's meaning, like "slavery" for "immigration." Why would Wikipedia, the map's training text, put it there?

**Share:** your word and that neighbor. Then: if the map had learned from censored text, what would change?



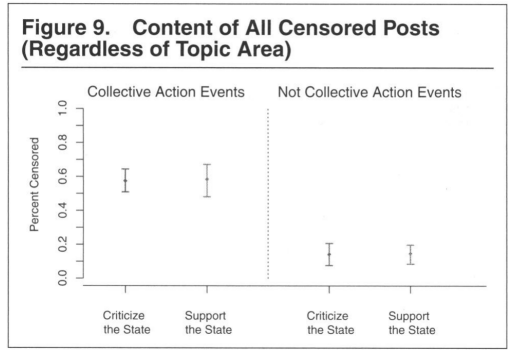

The words nearest "immigration."
A smaller number means closer.

Screenshot: Google's Embedding Projector, default data (Word2Vec 10K), October 6, 2026.

# Training text and politics

# What Chinese censors delete

- Researchers copied posts from nearly 1,400 Chinese sites before censors acted.
- Later they checked which posts were deleted, and compared the two piles across 85 topics.
- Criticism of the state, its leaders, or its policies: **no more likely** to be deleted than other posts.
- Posts about events that could bring people out together (protests, gatherings): deleted far more often, whatever they said about the state (about 80% on average in three such events).

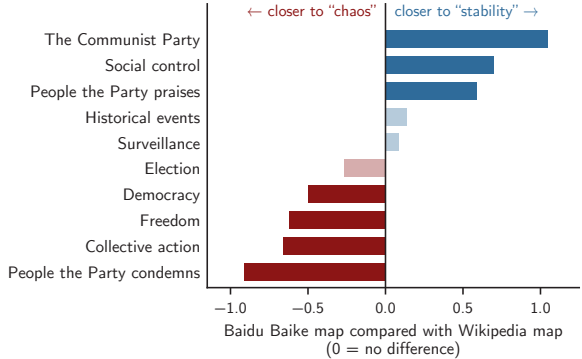



Share of posts deleted, all events. Criticism and support of the state are deleted at similar rates (no significant difference).

Source: King, Pan, and Roberts (2013), *American Political Science Review* 107(2), pp. 326 and 337; Figures 8 and 9, p. 337.

# Word embeddings from a censored encyclopedia

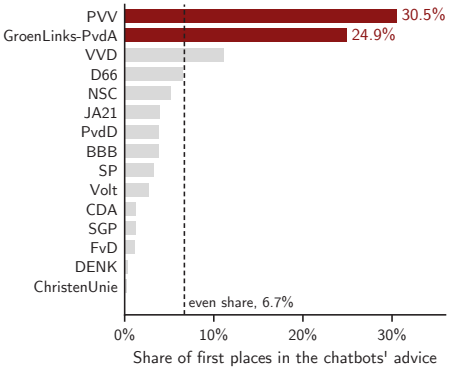
- Two training texts: Baidu Baike (censored, inside China) and Chinese Wikipedia (blocked in China). One word map learned from each.
- Each row is a group of words, such as words about democracy.
- On each map: does the group sit closer to "stability" words or to "chaos" words?
- Each bar: Baidu Baike's answer minus Wikipedia's.
  - Red: tilted more toward chaos than on Wikipedia's map.
  - Blue: tilted more toward stability.




Yang and Roberts (2021), Table 2. Bar length: a standardized effect size. Lighter bars: weaker evidence of a difference ($p > 0.05$).

# Chatbots as voting advice: the Netherlands, 2025

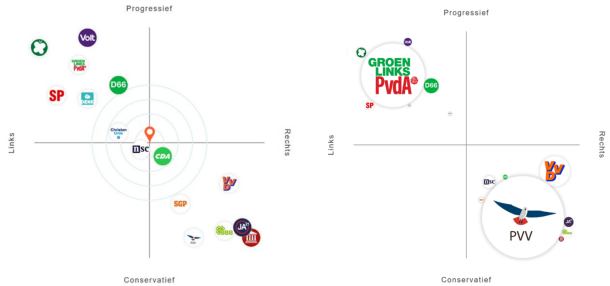
- Before the October 2025 election, the Dutch data-protection authority tested four chatbots: ChatGPT, Gemini, Mistral, and Grok.
- It made up 3,000 voters, 200 for each of the 15 parties in parliament. Each voter's answers to 30 statements copied one party's, with a few changed at random.
- It asked for the three parties that best fit each voter.
- With equal voters per party, each party should come first about 6.7% of the time.
- Two parties got 55% of the first places. Several got almost none.




Source: Autoriteit Persoonsgegevens (2025), "AI-chatbots als stemhulp," pp. 2 and 4 (Box 1), Figure 1, redrawn; eight smaller bars traced from the figure.

# Where the chatbots sent the voters

- Left: where a Dutch voting-advice website places the parties. Several sit near the center.
- Right: where the chatbots' advice landed. Most voters on the left were sent to GroenLinks-PvdA, most on the right to the PVV. The center almost vanished.
- Every chatbot showed this pattern. The authority: do not use chatbots for voting advice.




The parties (Kieskompas) The chatbots' advice

Axes: Links = left, Rechts = right, Progressief = progressive, Conservatief = conservative.

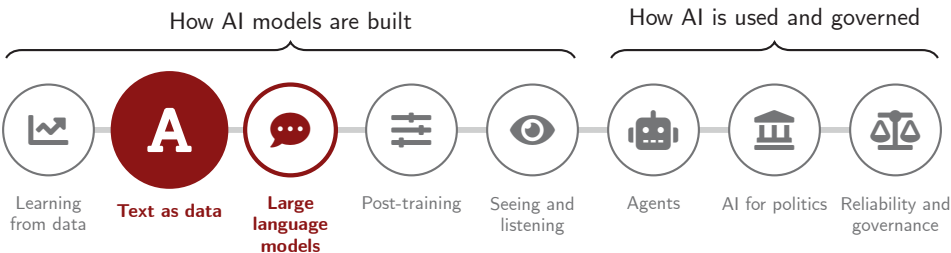
Source: Autoriteit Persoonsgegevens (2025), "AI-chatbots als stemhulp," pp. 3–4, Figures 3 and 4 (Figure 3 from Kieskompas).

# Discuss



1. What would an AI trained mostly on one country's text treat as normal? Give one example.
2. Which languages and communities are hard for AI to read, and who pays for that?
3. Back to today's question: if a model learned from censored text, how could you tell?

# Next week




**Next Wednesday, October 14**

Large language models.

# Backup slides

# A map of meaning

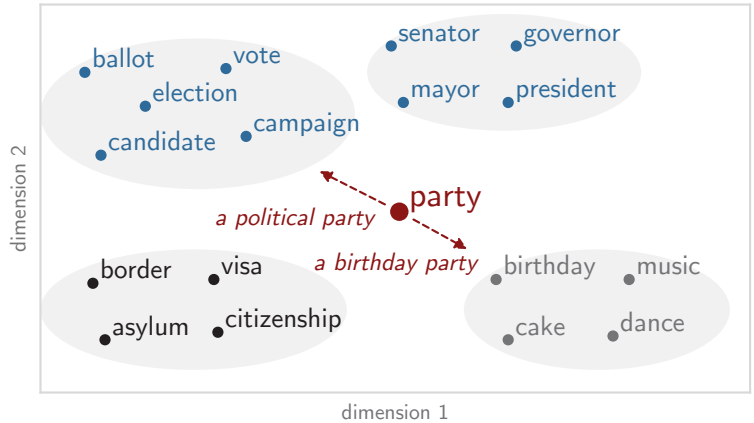



Illustration: positions placed by hand.

# Words that fit the same blanks

| Sentence with a blank | Words that fit |
|---|---|
| The bill was vetoed by the _____. | governor, president, mayor |
| Early this morning she cast her _____. | ballot, vote |
| The election was won by the _____. | party, candidate |
| Everyone danced at the _____. | party, wedding |

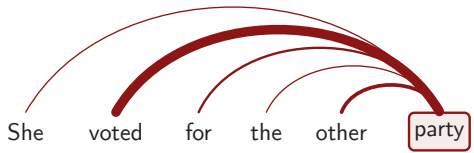
On the map: king $-$ man $+$ woman $\approx$ queen

Source: Mikolov, Chen, Corrado, and Dean (2013); Mikolov, Yih, and Zweig (2013); the 2025 course slides.

# One word, many meanings

| Word | One sentence | Another sentence |
|---|---|---|
| **party** | She voted for the other party. | We danced all night at the party. |
| **bank** | The bank raised its interest rates. | They sat on the bank of the river. |
| **right** | Voting is a right. | The right won the Senate. |

# Attention



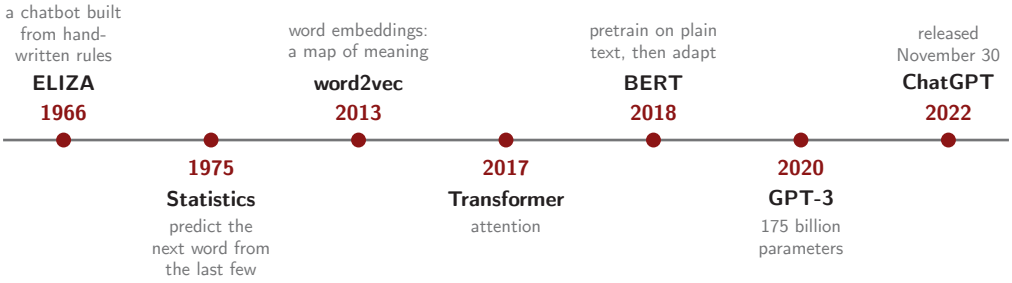

**party** reads as a political group

**party** reads as a celebration

- Line width: how much "party" draws on each earlier word.
- See real attention in your browser: poloclub.github.io/transformer-explainer

Illustration: line widths drawn by hand. In chatbot models, a word draws only on the words before it.

# From rules to large language models




Source: the 2025 course slides; Weizenbaum (1966); Jurafsky and Martin, ch. 3; Mikolov, Chen, et al. (2013); Vaswani et al. (2017); Devlin et al. (2019); Brown et al. (2020).

# Sources


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Prep: Financial Times (2023), "Generative AI exists because of the transformer"; 3Blue1Brown (2024), "Transformers, the tech behind LLMs".

Icons: Font Awesome Free 5 (CC BY 4.0). Video: Manim Community Edition; narration by Kokoro-82M, a synthetic voice; tokens from OpenAI's Whisper tokenizer, run locally.